

# Quantifying uncertainty when extrapolating the relationship between snorkel counts and mark-recapture estimates of juvenile salmonids

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## Abstract

Snorkel surveys are frequently used to monitor stream-dwelling fish. Inferring local abundance from snorkel surveys is complicated by two primary factors: variable fish detection probabilities and the relative abundance of fish in habitat types below the recommended minimum depth for snorkeling. We examine these factors across three salmonid species (*Oncorhynchus* spp.), 4 years, and 113 location-years in Oregon coastal streams. We calibrate snorkel counts to mark-recapture estimates and develop mathematically explicit expressions that convert a new snorkel count into a probability density of abundance for streams containing mixed habitat types that were and were not snorkeled. Snorkelers detected 63%, 47%, and 39% of juvenile coho salmon (*Oncorhynchus kisutch*), steelhead trout (*Oncorhynchus mykiss*), and cutthroat trout (*Oncorhynchus clarkii clarkia*) estimated by mark-recapture, respectively, but uncertainty within and among sampling units was high. Further, analytics developed here can be used to infer abundance and uncertainty for habitat types that were not snorkeled. Our quantification of uncertainty arising from using snorkel counts as a proxy for abundance will help managers balance biological risks with available resources.

**Key words:** salmonid monitoring, snorkel surveys, Bayesian mark-recapture, density-dependent habitat selection

## Introduction

Snorkel surveys are widely used to monitor juvenile salmonid abundance (Crawford 2011; Jonasson et al. 2016; Poole et al. 2019). The purpose of snorkel surveys ranges from the evaluation of recovery efforts (e.g., McElhany et al. 2000; Flitcroft et al. 2012, 2014; O'Neal et al. 2016) to examination of ecological relationships (Hicks and Hall 2003; May and Lee 2004). Snorkel surveys are often performed because they cost less than electrofishing or mark-recapture and thus may be conducted with greater replication in time and/or space (Schill and Griffith 1984; Hankin and Reeves 1988; Dolloff et al. 1993). Furthermore, snorkeling can be conducted in habitats that are too deep or complex to effectively sample with techniques that require fish to be captured (Gardiner 1984; Hicks and Watson 1985). Snorkeling also poses less risk of mortality than methods that require capture, making it well suited for monitoring species listed under the Endangered Species Act (ESA; O'Neal 2007).

Inference from snorkel counts is confounded by two primary factors. First, the detection probability of snorkeling is often unknown (e.g., Hankin and Reeves 1988; Walters et al. 1999), or can vary with differing survey types and species (Staton et al. 2022), environmental conditions (Hayes and Baird 1994; Mullner et al. 1998), and fish densities (Hillman

et al. 1992; Roni and Fayram 2000). Second, a depth of at least 20 cm is recommended for snorkeling (O'Neal 2007). The proportion of fish distributed into habitats below this recommended depth is often unknown and can exhibit inter-annual variation, complicating an attempt to infer local abundance within a reach using detection-adjusted snorkel counts (Constable and Suring 2023). Previous attempts to quantify variation in abundance between sample units that can and cannot be snorkeled are rare.

The need to address these factors was motivated by management concerns in Western Oregon, where coho salmon (*Oncorhynchus kisutch*) are listed under the ESA and co-occur with steelhead (anadromous *O. mykiss*) and cutthroat trout (*O. clarkii clarkii*). In response to the ESA listing, a large-scale monitoring effort was initiated in 1998 under the Oregon Plan for Salmon and Watersheds (OPSW) that yields snorkel counts of all three species (State of Oregon 1997). The OPSW monitoring area includes 16 700 km of streams in three coho salmon Evolutionarily Significant Units and four steelhead Distinct Population Segments (Fig. S6). Snorkeling was used to monitor this area because of its increased spatial coverage relative to other methods. Snorkeling was also favorable because coho salmon prefer pools (Nickelson et al. 1992; Bisson et al. 1997), which typically provide adequate depths for snorke-

ling but can be too deep or complex to efficiently sample for methods that require fish capture (Mullner et al. 1998; Satterthwaite 2002). Further, stream visibility and temperatures in this area were conducive to snorkeling (Hillman et al. 1992; Mullner et al. 1998; Roni and Fayram 2000). Under the OPSW protocols, snorkeling is only conducted in pools that are  $\geq 20$  cm deep. Other protocols similarly sample only in pools (e.g., Flitcoft et al. 2014; Tualatin River Watershed Council 2015) or pools and glides (e.g., Ebersole et al. 2009).

The OPSW monitoring data are used to assess trends in the distribution and abundance of salmonids (Constable and Suring 2022) and inform the status of coho salmon (Wainwright et al. 2008; Stout et al. 2012). For trend monitoring, a method should be sufficiently precise and have consistent bias (Temple and Pearsons 2007). Counts from snorkel surveys have demonstrated repeatability (Thurrow 1994). Resurveys of OPSW snorkel sites have indicated that snorkel counts are precise (Constable and Suring 2022) but less was known about snorkel survey bias. The magnitude of snorkeler detection probability was previously thought to change according to visibility conditions (Rodgers et al. 1992), species, life stage, and number of fish (Hillman et al. 1992). Several studies have compared snorkel counts to electrofishing removal estimates (e.g., Roni and Fayram 2000; Ebersole et al. 2009). However, electrofishing estimates also have variable biases related to both fish and stream attributes (Peterson et al. 2004; Rosenberger and Dunham 2005). Until now, little has been done to compare the accuracy snorkel counts of coho salmon with known numbers of fish or methods with less bias, such as mark-recapture, since the assessments of Rodgers et al. (1992) and Hillman et al. (1992). The accuracy of OPSW steelhead and cutthroat counts is unknown, and we are unaware of any attempt to infer uncertainties in abundance in fast-water habitats (e.g., riffles and rapids) for these three species given snorkel counts in pools for any protocol.

Here, we address the two factors that confound inference from snorkel counts by developing techniques to convert a new snorkel count into a probability distribution of abundance in stream reaches containing a mix of habitats that were and were not snorkeled. From 2016 to 2019, we sampled 18 stream reaches (Supplement 1). In 66 pools, both snorkel and mark-recapture estimates were conducted. Mark-recapture estimates were conducted in an additional 47 fast-water units. We develop novel analytics to quantify uncertainty of fish abundance in pool habitats that were snorkeled and in fast-water units where snorkeling was not conducted.

## Materials and methods

### Ongoing OPSW monitoring

The OPSW monitoring protocol employs a generalized random tessellation stratified design to select random, spatially balanced sites that are representative of the sampling frame (Stevens and Olsen 2004). The OPSW sampling frame includes all first- to third-order streams within coho salmon and steelhead rearing distributions. Selected sites were surveyed using a snorkeling protocol developed by Rodgers (2000) and a stream habitat survey protocol developed by Moore et al.

(1997). Surveying took place during daylight hours of the summer base flow period (mid-July to early October). Snorkelers began at the downstream end of the site and proceeded upstream (Thurrow 1994). The type of each habitat unit was determined following Moore et al. (1997). Fast-water units had slopes  $> 0$ , turbulent surface water, and were more uniform in channel bed form, relative to pools. Pools had a slope of 0, still surface water, and contained a well-defined scour within the channel bed form or an impounding feature, such as a beaver dam or log jam. The length, width, and depth of each fast-water unit (e.g., riffles, rapids, cascades) and pool was measured. Fast-water units were not snorkeled under OPSW protocols because the majority (78%, Constable, Oregon Department of Fish and Wildlife, unpublished data) of these habitats in the program's sampling frame did not meet the recommended 20 cm minimum depth criteria. Only pools that were  $\geq 20$  cm in maximum depth and  $\geq 6$  m<sup>2</sup> in surface area were snorkeled. Pools meeting these criteria were surveyed in a single snorkeling pass, during which snorkelers made separate counts of cutthroat trout and juvenile steelhead  $\geq 90$  mm fork length (FL), and juvenile coho salmon regardless of length. Juvenile steelhead and cutthroat trout  $< 90$  mm FL were not counted because they cannot be reliably distinguished by snorkelers in this size range (Hawkins 1997; Roni and Fayram 2000). Steelhead adults were rarely observed during snorkel surveys because spawning run timing does not coincide with the survey period (reviewed by Busby et al. 1996). Resident *O. mykiss* (rainbow trout) are also rare in OPSW project area, occurring mainly in habitats that are upstream of barriers to anadromy (Busby et al. 1996), which are outside of the sampling frame. In most years, 10%–15% of surveyed sites were resurveyed to determine adherence to survey protocols and the count precision.

### Data collection

In our study, streams containing juvenile coho salmon, juvenile steelhead, and cutthroat trout from past OPSW surveys were chosen to conduct concurrent snorkel counts and mark-recapture estimates. Within chosen streams, a reach was identified that contained four pool habitat sampling units (hereafter “pool”) with fast-water units between each pool. Thus, four pools and three fast-water units could be surveyed in sequence. Pools were required meet the OPSW depth and surface area criteria for snorkeling. Once this reach was located, block nets were placed to establish closed populations (Peterson et al. 2005). Block nets were placed at the top and bottom of the reach and at the transition between each pool and fast-water unit to prevent fish from moving between sampling units. The integrity of each block net was examined by a snorkeler. This configuration allowed for a snorkel count in each pool and a mark-recapture data collection in each pool and in each fast-water unit. Though block nets are not typically present during snorkeling, we assumed their presence would not significantly bias snorkel counts. This assumption was based on OPSW field observations and the work of Korman et al. (2010), who observed steelhead remaining stationary for visual observations. Additionally, mark-resight studies with observations find few fish from the

focal reach in adjacent reaches the day after marking (Tim Copeland, Idaho Fish and Game, personal communication).

Snorkel counts preceded mark-recapture estimates. Snorkel counts were made following OPSW protocols described above (Rodgers 2000). Visibility and hiding cover were rated on an ordinal scale by the snorkeler, considering water clarity and any other factors that could impede the ability to observe fish (Rodgers 2000; Crawford 2011). After snorkeling was completed, the reach was allowed to recover 2–2.5 h before the first phase of mark-recapture began. Fish were captured by electrofishing with 1–3 Smith Root, Inc. model 12-B backpack electrofishers. Mode switches were set at G-5 (40 Hz at a 25% duty cycle) and voltage settings ranged from 100 to 300 V; depending on stream depth, temperature, conductivity, and fish response. Each pool and fast-water was electrofished until the entire area of the pool was covered. Fish were identified to species, counted, and marked for recapture. Fish captured in pool units were marked differently than fish captured in fast-water units. When captured fish had recovered and when electrofishing and marking was completed in the subsequent upstream habitat unit, fish were released back into the unit where they were captured. Fish recaptured from outside the unit in which they were marked were not included in mark-recapture estimates. Mortalities recorded in the marking phase were not available for recapture and were not included in mark-recapture estimates. Mortalities recorded in the recapture phase were included in mark-recapture estimates.

When the marking phase was complete, the reach was allowed to recover ~24 h before the recapture phase began (Temple and Pearsons 2007). The duration of this recovery period should satisfy our subsequent mark-recapture modeling assumption that capture efficiency was similar between passes (Peterson et al. 2004; Rosenberger and Dunham 2005) and similar among marked and unmarked individuals. Block nets remained in place during this time and were inspected periodically and cleaned of debris. Each pool and fast-water unit was electrofished using similar effort as the first phase, covering all areas of the habitat unit with a goal of recovering approximately 50% or more of the marked fish (Rodgers et al. 1992). Numbers of marked and unmarked fish were recorded for each species and mark location was recorded to determine if fish had migrated past block nets (violating mark-recapture assumptions). As with the marking phase, fish were held in buckets within the unit where they were captured and released when they had recovered and when electrofishing was completed in the subsequent upstream unit.

## Mark-recapture analysis

There are four possible capture histories for an individual sampled on two occasions: {1,1}, {1,0}, {0,1}, and {0,0}, where 1 indicates that the individual is observed, 0 indicates that it was unobserved, and commas within a capture history {} separate first and second sampling occasions. A simple model for these data assumes that capture probability,  $p$ , is a constant for all individuals within a sampling unit, and constant across sampling occasions (but not necessarily constant across sampling units). Thus, under the constant  $p$  assumption, the four

possible capture histories listed above should occur with frequency  $p^2$ ,  $p(1-p)$ ,  $(1-p)p$ , and  $(1-p)^2$ , respectively. The foregoing can be further simplified into just three kinds of observations. An individual can be observed on both occasions,  $x = 2$ , with probability  $p^2$ ; observed once,  $x = 1$ , with probability  $2p(1-p)$ ; or never observed,  $x = 0$ , with probability  $(1-p)^2$ . These arithmetic facts are also given by the binomial distribution

$$(1) \quad f(x; 2, p) = \binom{2}{x} p^x (1-p)^{2-x}$$

where  $x$  is 0, 1, or 2, indicating the number of times an individual was observed. The number 2 in eq. 1 indicates two sampling occasions in which individuals are available for observation. Finding the value of  $p$  that maximizes the likelihood of data  $x$  is problematic because the number of individuals with capture history {0,0} is axiomatically unknown and so the number of instances in which  $x = 0$  is thus unavailable. Huggins (1991) addressed this issue for maximum likelihood analyses by developing a likelihood that conditions estimation of  $p$  only on observable capture histories.

Data augmentation with Bayesian techniques also provides a work-around to the problem of unknown number of individuals with the {0,0} capture history. With data augmentation (Royle et al. 2007), we add an arbitrary number of “synthetic” individuals with capture history {0,0} to the observed data set. The analytical task becomes estimating the proportion of real animals in the augmented data set. To do this, we introduce a parameter,  $\psi$ , for the inclusion probability that a given individual in the augmented data set is a real individual. For all individuals  $i$  in the augmented data set we have

$$(2) \quad z_i \sim \text{Bernoulli}(\psi)$$

$$(3) \quad x_i \sim \text{binomial}(p \cdot z_i, \text{trials} = 2)$$

and the sum over  $z$  is the estimate of abundance (Kéry and Schaub 2012). We augmented all species-and-unit data sets with 100 synthetic individuals.

Here, we extend the data augmentation technique to accommodate our multiple sampling units ( $j$ ), each having unique covariates and observed snorkel counts. The inclusion probability of synthetic individuals was assumed to vary across units, and flat prior distribution was used:  $\psi_j \sim \text{Beta}(1,1)$ . We checked that posteriors of  $\psi$  were not truncated at 1 to verify that data were sufficiently augmented. Three alternative models are

$$(4) \quad \text{Model1} : p_j \sim \text{beta}(1, 1)$$

$$(5) \quad \text{Model2} : p_{\text{fast}} \sim \text{beta}(1, 1); p_{\text{pool}} \sim \text{beta}(1, 1)$$

$$(6) \quad \text{Model3} : \text{logit}(p_j) = b_0 + b_1 * \text{Clarity} + b_2 * \text{Cover}; b. \sim \text{Normal}(\mu = 0, \sigma = 32)$$

where the distributions stated within each model are Bayesian prior distributions. The large standard deviation ( $\sigma$

**Table 1.** Two information-theoretic criteria (eviance information criterion (DIC) and atanabe–Akaike information criterion (WAIC)) were calculated separately for each species and model type (1–3 vs. 1b–4b).

|       |    | Coho  |        |      | Steelhead |        |      | Cutthroat |        |      |
|-------|----|-------|--------|------|-----------|--------|------|-----------|--------|------|
|       |    | Δ DIC | Δ WAIC | GoF  | Δ DIC     | Δ WAIC | GoF  | Δ DIC     | Δ WAIC | GOF  |
| Model | 1  | 1458  | 423    | 0.23 | 0         | 403    | 0.27 | 0         | 380    | 0.27 |
|       | 2  | 49    | 23     | 0.36 | 1990      | 18     | 0.37 | 894       | 0      | 0.39 |
|       | 3  | 0     | 0      | 0.36 | 1220      | 0      | 0.39 | 1100      | 15     | 0.37 |
|       | 1b | 1547  | 392    | 0.31 | 158       | 319    | 0.52 | 0         | 274    | 0.50 |
|       | 2b | 0     | 44     | 0.42 | 1659      | 66     | 0.62 | 627       | 26     | 0.60 |
|       | 3b | 184   | 0      | 0.45 | 0         | 0      | 0.70 | 201       | 0      | 0.63 |
|       | 4b | 782   | 492    | 0.25 | 315       | 340    | 0.37 | 337       | 305    | 0.37 |

Note: GoF is goodness of fit as described in eqs. 12 and 13 of main text. The R code for generating these metrics is given in Supplement 6.

= 32) is equivalent to a vague precision of 0.001. Note that Model 1 estimates  $p$  uniquely for each of the  $j$  sites; Model 2 estimates  $p$  twice for sites dichotomously classified as either “fast” or “pool”; Model 3 derives  $p$  for each of the  $j$  units using a logit-linear model with the covariates *clarity* and *cover*. Thus, in Model 3, the value of  $p$  at unit  $j$  depends on the across-unit relationship between  $p$  and covariates *clarity* and *cover*.

A posterior distribution for the proportion of the local abundance at a site that is observed by a snorkeler is obtained simply by defining the quantity

$$(7) \quad p_{\text{snorkel},j} = \frac{\text{Observed}_{\text{snorkel},j}}{\hat{N}_j}$$

and then including this node in the Markov chain Monte Carlo (MCMC) algorithm. The other derived quantities, mean abundances across sites on different years, mean abundances across years in different habitat types (pool vs. fast) and mean ratios of abundance in fast vs pool sampling units for different years can be similarly estimated with uncertainty inside the MCMC algorithm without needing to specify additional priors.

If the snorkel count is informative of abundance, then the snorkel count data can be integrated directly into the abundance estimate. Rather than estimating abundance across sites and then quantifying the relationship with the associated snorkel counts, as in eq. 7, it is also possible to leverage a potential relationship when estimating abundance at each site. Specifically, we can replace eq. 7 with

$$(8) \quad \text{Observed}_{\text{snorkel},j} \sim \text{Binomial}(p_{\text{snorkel},j}, \hat{N}_j)$$

In this case, we estimate abundance  $\hat{N}_j$  jointly in eq. 9 and over the sum of  $z$  in the data augmentation. This joint estimation of abundance from two different data sources is known as an integrated model (Schaub and Abadi 2011). In Table 1 and throughout, models 1, 2, and 3 that are fitted using eq. 9 instead of eq. 7 (those that integrate the snorkel counts) are denoted with the letter “b”.

In Model 3, we ask whether mark-recapture capture probability is related to the stream habitat covariates *clarity* and *cover*. We may also ask whether these covariates affect snorkel detection probability. This is done with:

$$(9) \quad \text{Logit}(p_{\text{snorkel},j}) = \gamma_0 + \gamma_1 * \text{Clarity} + \gamma_2 * \text{Cover}; \quad \gamma_i \sim \text{Normal}(\mu = 0, \sigma = 32)$$

For simplicity, using stream covariates on snorkel detection probability was implemented only when the mark-recapture capture probability was treated as a unique random variable in each mark-recapture study (Model 1 above). We call this Model 4b below.

Model comparisons were made with two different information-theoretic metrics. The deviance information criterion (DIC; Spiegelhalter et al. 2002) computed with package R2jags uses half the variance of the deviance as an approximation for the effective number of parameters. We also calculated Watanabe–Akaike information criterion (WAIC). This was done by recording the point-wise predictive density of each datum  $x_i$  in a species-specific data set of size  $N$ ,

$$(10) \quad D_i = \text{binomial}(x_i, p \cdot z_i, \text{trials} = 2)$$

across all iterations,  $s$ , of the MCMC, yielding  $D_{i,s}$ . Following Gelman et al. (2014), we computed WAIC from the posterior samples after completing the MCMC:

$$(11) \quad \text{WAIC} = -2 \sum_{i=1}^N \log \left( \frac{1}{S} \sum_{s=1}^S D_{i,s} \right) - \sum_{i=1}^N \frac{1}{S-1} \sum_{s=1}^S \left( \log(D_{i,s}) - \log(\bar{D}_i) \right)^2$$

The “b” versions of the models contain two densities.

Model goodness of fit (GoF) was assessed with a posterior predictive check (Bayesian  $p$ -value, Gelman et al. 1996). For all real and augmentation individuals,  $i$ , we recorded Freeman–Tukey (FT) statistics

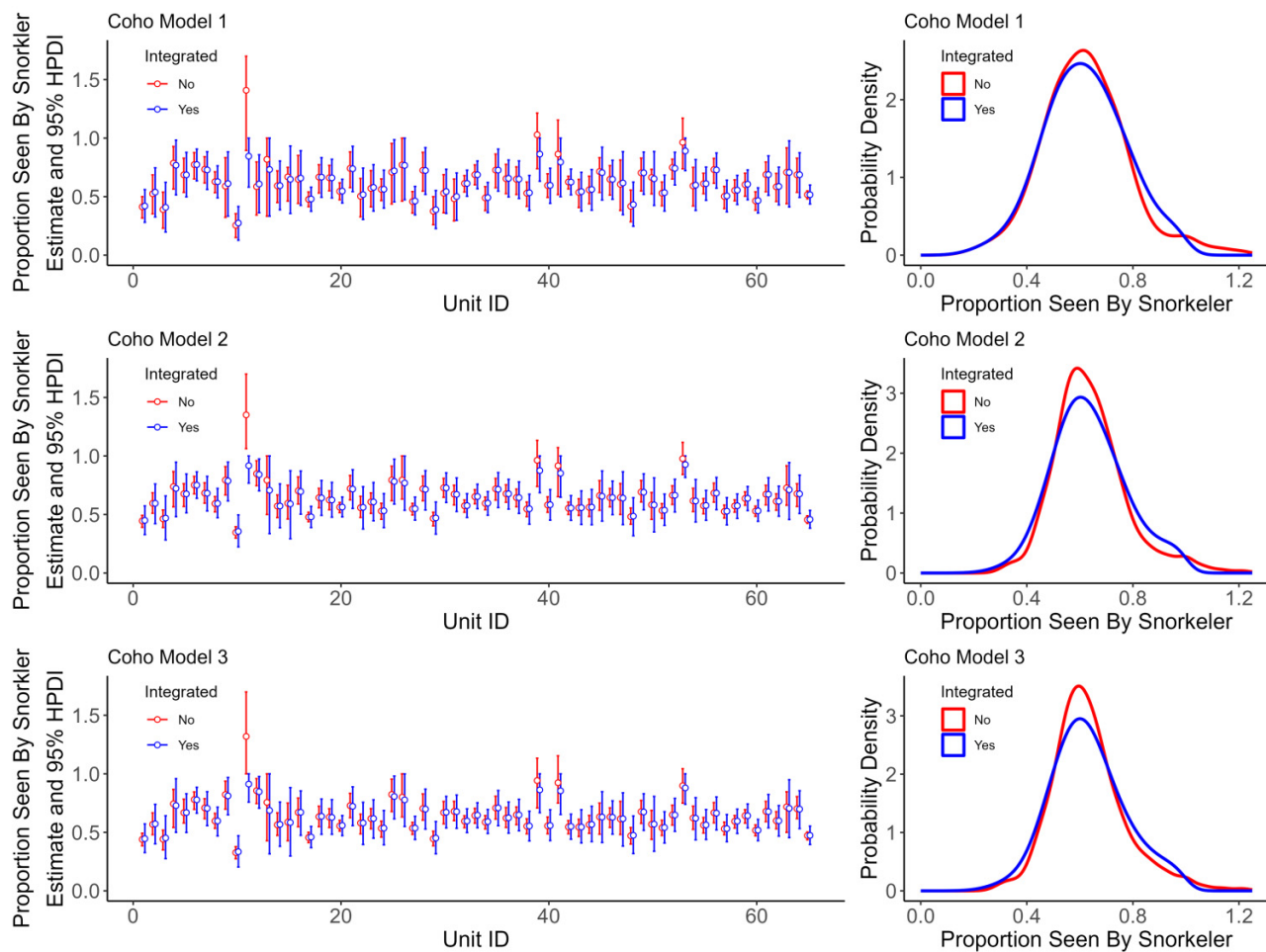
$$(12) \quad \text{FT.obs}_i = \left( \sqrt{x_i} - \sqrt{E(x_i)} \right)^2$$

$$(13) \quad \text{FT.sim}_i = \left( \sqrt{x.\text{sim}_i} - \sqrt{E(x_i)} \right)^2$$

at each iteration of the MCMC, where  $E(x_i) = p_i * z_i * 2$  and  $x.\text{sim}_i \sim \text{Binomial}(p_i * z_i, \text{size} = 2)$ . We summed  $\text{FT.obs}$  and  $\text{FT.sim}$  over individuals and checked for the proportion of MCMC iterations where  $\text{FT.sim} > \text{FT.obs}$ .

Models were implemented with R Statistical Software (V4.2.3; R Core Team 2023) and JAGS (V4.3.0; Plummer 2003). For each model formulation-by-species cross ( $7 * 3 = 21$  models), we ran 50 000 iterations of three MCMC simulations with

**Fig. 1.** Point estimates and 95% highest probability density intervals (HPDI) for the proportion of juvenile coho seen by snorkelers at each pool unit where mark-recapture and snorkel counts were conducted concurrently (left column). The integrated models in blue include the snorkel count in the estimation of local abundance (denoted “b” in the main text). Alternative model formulations (rows) produce similar results. Aggregating uncertainty within and across sampling unit results in the probability density distributions for the proportion of fish seen by snorkelers given in the right-hand column.



a 15 000 iteration burn-in period. Chains were thinned to 1:5, resulting in 21 000 posterior samples per parameter per model per species. A high-performance computer server at the University of Idaho was needed to prevent memory overload and had an additional benefit of running all 63 MCMC chains simultaneously.

### Inferring abundance in pool sample units

We desire a probability density function for the abundance of fish in a pool given a new snorkel count in a pool. This is possible using our empirical estimates of the proportion seen by snorkelers. Denoting a new snorkel count as SC and the now estimated proportion seen by snorkelers as PS, our estimator of abundance in the pool is

$$(14) \quad \widehat{N}_{\text{pool}} = \frac{\text{SC}}{\text{PS}}$$

We assume the new snorkel count comes from a pool unit drawn randomly from the collection of pools sampled for this study. We now develop a probability density function for  $\widehat{N}_{\text{pool}}$  that includes uncertainty arising from random sampling of

pool unit as well as uncertainty in the proportion of fish seen by snorkelers. Uncertainty in the proportion of fish seen by snorkelers across all units is given in the right-hand column of Figs. 1–3. We note right skew in these posterior distributions arising from a few sites where snorkelers observed a relatively high proportion of available fish. We therefore selected a gamma distribution to generalize the empirical posterior distribution of the among-sites proportion of fish seen by snorkelers. Given that we have a gamma distribution in the denominator of eq. 14, it is not trivial to find the density function for abundance in the pool given some new snorkel count. The change-of-variables technique provides the solution. Let  $w()$  denote the inverse function, and  $w'()$  be the derivative of the inverse function. The change-of-variables technique stipulates that

$$(15) \quad g(\widehat{N}_{\text{pool}}) = f(w(\widehat{N}_{\text{pool}})) * |w'(\widehat{N}_{\text{pool}})|$$

Noting that the inverse function of eq. 14 is

$$(16) \quad w(\widehat{N}_{\text{pool}}) = \text{PS} = \frac{\text{SC}}{\widehat{N}_{\text{pool}}}$$

**Fig. 2.** Mean unit abundance (x-axis) for each of three species (panels) and four years (y-axis). All estimates were generated with Model 3b.



and  $f()$  is the gamma function parameterized with shape  $\alpha$  and rate  $\beta$ , we obtain

$$(17) \quad g(\widehat{N}_{pool}; sc, \alpha, \beta) = \frac{\widehat{N}_{pool}^{\alpha-1} e^{-\beta \frac{sc}{\widehat{N}_{pool}}} \beta^\alpha}{(\alpha-1)!} * \left| -\frac{SC}{\widehat{N}_{pool}^2} \right|.$$

To implement, we find values for  $\alpha$  and  $\beta$  that maximize the likelihood of our MCMC posterior samples for the species of interest. These values are inserted into eq. 16. We may also insert any value for new snorkel count SC and obtain

$$(18) \quad \int_0^\infty \frac{\widehat{N}_{pool}^{\alpha-1} e^{-\beta \frac{sc}{\widehat{N}_{pool}}} \beta^\alpha}{(\alpha-1)!} * \left| -\frac{SC}{\widehat{N}_{pool}^2} \right| d\widehat{N}_{pool} = 1,$$

indicating that we have a proper posterior probability. From here, we can use numerical procedures to find the 95% highest probability density interval (HPDI) for  $\widehat{N}_{pool}$ . This is done by numerically integrating eq. 18 across variable ranges of the

x-axis ( $\widehat{N}_{pool}$ ) and retaining the range integrating to 0.95 that is higher on the y-axis (probability density) than all other 0.95 integral values.

## Inferring abundance in fast sample units

We desire a probability density function for the abundance of fish in a fast unit given a new snorkel count in a pool unit. This is possible using our empirical estimates of the proportion seen by snorkelers (PS) and the empirical estimates of the ratio of abundance in fast and pool units (*RatioFP*) which was calculated as a derived parameter in the MCMC procedure.

We begin by noting that

$$(19) \quad \widehat{N}_{fast} = \frac{SC}{PS} * RatioFP$$

where the first term on the right-hand side is the pool abundance (eq. 14). We pool MCMC samples of *RatioFP* across all years and then once again fit a gamma distribution using maximum likelihood. This yields a closed-form parametric model for the second term on the right-hand side of eq. 19. Since we already have a closed-form solution for the first term on the right side of eq. 19 (viz. eq. 17), we could proceed with known methods for calculating a product distribution. However, a more straightforward approach follows from noting that the distribution of the quotient of two gamma-distributed random variables is beta-prime distributed

$$(20) \quad \frac{RatioFP}{PS} \sim \frac{(x/\sigma)^{\alpha-1} (1+x/\sigma)^{-\alpha-\beta}}{B(\alpha, \beta)}$$

where  $B$  is the Beta function,  $\alpha$  and  $\beta$  are the shape and rate parameters, respectively, of *RatioFP*, and  $\sigma$  is the rate parameter of PS divided by the rate parameter of *RatioFP*.

The parametric form in eq. 20 is multiplied by a constant (SC) to obtain a posterior for  $\widehat{N}_{fast}$ . We employ the change-of-variables technique again to obtain

$$(21) \quad h(\widehat{N}_{fast}; SC, \alpha, \beta, \sigma) = \frac{\left(\frac{\widehat{N}_{fast}}{SC\sigma}\right)^{\alpha-1} \left(1 + \frac{\widehat{N}_{fast}}{SC\sigma}\right)^{-\alpha-\beta}}{B(\alpha, \beta)} * \left| \frac{1}{SC} \right|$$

This solution for the posterior distribution of  $\widehat{N}_{fast}$  can be confirmed with Monte Carlo techniques (Appendix A). With this closed-form posterior for  $\widehat{N}_{fast}$  we can compute 95% HPDI for any given snorkel count in pool sample units.

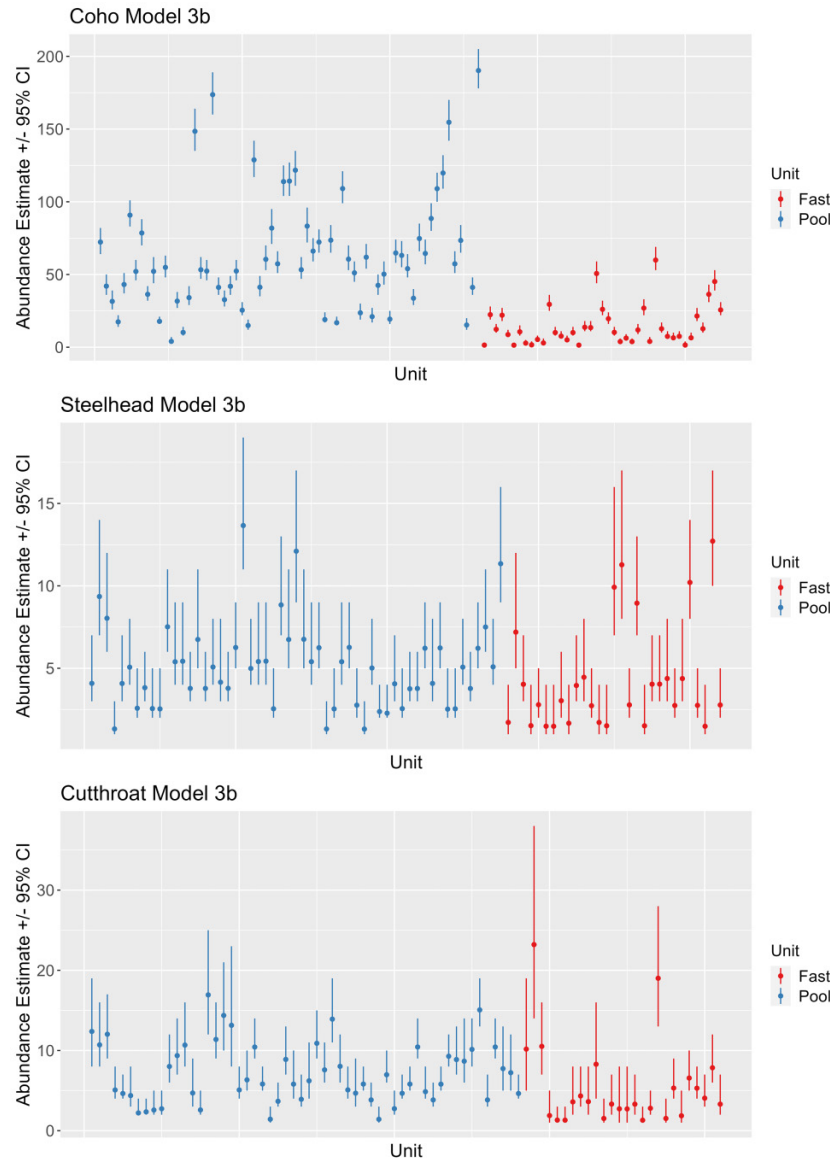
## Density-dependent snorkel detection

Our mark-recapture models make no assumption about the relationship between pool sampling unit abundance and snorkel detection probability. To determine whether snorkel detection varies with abundance, we regress year-by-pool point estimates of the proportion seen by snorkelers,  $p_{snorkel}$ , against the associated point estimate of abundance for that pool and year. This is repeated for each species' Model 1, 1b, 2, 2b, 3, and 3b.

## Density-dependent habitat selection

Density-dependent habitat selection could cause the relative abundance of fish in fast and pool sampling units to change as site-level abundance changes across space or time. Such a response would add complexity when using snorkel

**Fig. 3.** Point estimates and 95% highest probability density intervals (HPDI) for three species (rows) in pool (blue) and fast-water (red) units. All estimates were generated with Model 3b.



counts (collected in pools) to infer abundance in fast sampling units or even site-level abundance. Simply plotting the abundance ratio (Fast:Pool) over site abundance is confounded by different sites having different numbers of sampling units in fast and pool sampling units. To remove the effect of different proportion of fast-water versus pool units within a reach, we calculate the mean abundance ratio of Fast:Pool for a given site and year ( $y$ -axis) as

$$\frac{\frac{1}{F} \sum_{f=1}^F N_f}{\frac{1}{P} \sum_{p=1}^P N_p}$$

where  $N$  is all posterior samples of abundance in fast ( $f$ ) and pool ( $p$ ) sampling unit. Aggregating the posterior samples across sampling units enables calculation of a 95% HPDI. We plot this over the mean unit abundance per site,

$$\frac{1}{F+P} \sum_i^{F+P} N_i$$

where  $N_i$  are posterior samples of abundance across all units  $i$  within the given site and year, regardless of the number of units within that site and year that are fast ( $F$ ) and pool ( $P$ ). We report the standard error of the point estimates of abundance among all units (fast and pool) within the site. We used Type I and Type II regression to test for an effect of mean unit abundance per site on the Fast:Pool ratio to determine if site-level abundance affects the distribution of fish between pool and fast-water units. We also explored using the inverse of the range of the 95% envelope of the Fast:Pool ratio in a weighted regression.

## Results

Pool and fast-water unit specific fin clips indicated 0.2% of coho salmon, 0% of steelhead, and 1.8% of cutthroat trout were recaptured outside of the unit in which they

were marked, suggesting block nets contained fish within their original sampling unit. Of all fish handled for mark-recapture, 0.7% of coho salmon, 0.8% of steelhead, and 0.3% of cutthroat trout were electrofishing related mortalities. These data indicated closed population assumptions for mark-recapture were met. Snorkelers did not observe fish crowding at block nets, suggesting that fish did not attempt to escape pools during snorkeling.

Model convergence was verified by visually inspecting trace plots. The maximum Gelman and Rubin (1992) convergence diagnostic Rhat across all 6384 monitored parameters was less than 1.05. The 10th, 50th, and 90th percentiles of the effective number of samples per parameter were 820, 11 000, and 21 000, respectively. No mark recapture model formulation consistently outperformed others when looking across species and information theoretic criteria (Table 1). However, there were generally small differences in parameter estimates across mark recapture models for coho (Fig. 1) as well as steelhead (Fig. S1) and cutthroat trout (Fig. S2). All models had satisfactory goodness of fit (Table 1). For simplicity, we hereafter highlight Model 3b because it had good information theoretic and goodness of fit metrics. Supplement 1 contains percentiles of posteriors for all parameters, as well as the associated Rhat and effective MCMC sample sizes for all models and species.

Integrating the snorkel count into the estimate of abundance did not systematically affect the abundance estimate (Figs. 1, S3, and S4). However, as noted above, units with sparse mark-recapture data had poor goodness of fit unless the snorkel count was integrated into the abundance estimate. Generalizing the proportion of fish seen by snorkelers across sample units is given in the right-hand columns of Fig. 1, Figs. S3, and S4. Using Model 3b, the mean of these distributions for coho, steelhead, and cutthroat trout are 63%, 47%, and 39%, respectively. However, uncertainty within and across sample units was high.

The mean unit abundance varied considerably across years for coho, with higher abundances on odd years (Fig. 2). In contrast, steelhead and cutthroat trout displayed more consistent mean unit abundances across the four years of study (Fig. 2). On average, pool sample units contained 3.3 times more coho than fast-water units (Fig. 3). Steelhead had equitable abundances in pool and fast-water units, but uncertainty in all estimates was high (Fig. 3). Cutthroat trout had slightly higher abundances in pools than fast-water units, and there was more variation in abundance across pool units than fast-water units (Fig. 3). The proportion of coho residing in pools was greater than that of steelhead and cutthroat trout (Fig. 4). All three species exhibited some interannual differences in the fast:pool abundance ratio, but the magnitude of annual differences was less than the differences among species (Fig. 4).

The mean unit abundance at a site consisting of multiple sample units did not predict the ratio of abundances among pool and fast-water units (Fig. 5). Type I and Type II linear regressions fit to these points were not significant at  $\alpha = 0.05$ . Using the inverse of the range of the 95% HPDI as weights in a linear regression similarly yielded insignificant effects of site abundance on the ratio of abundances among the site's pool

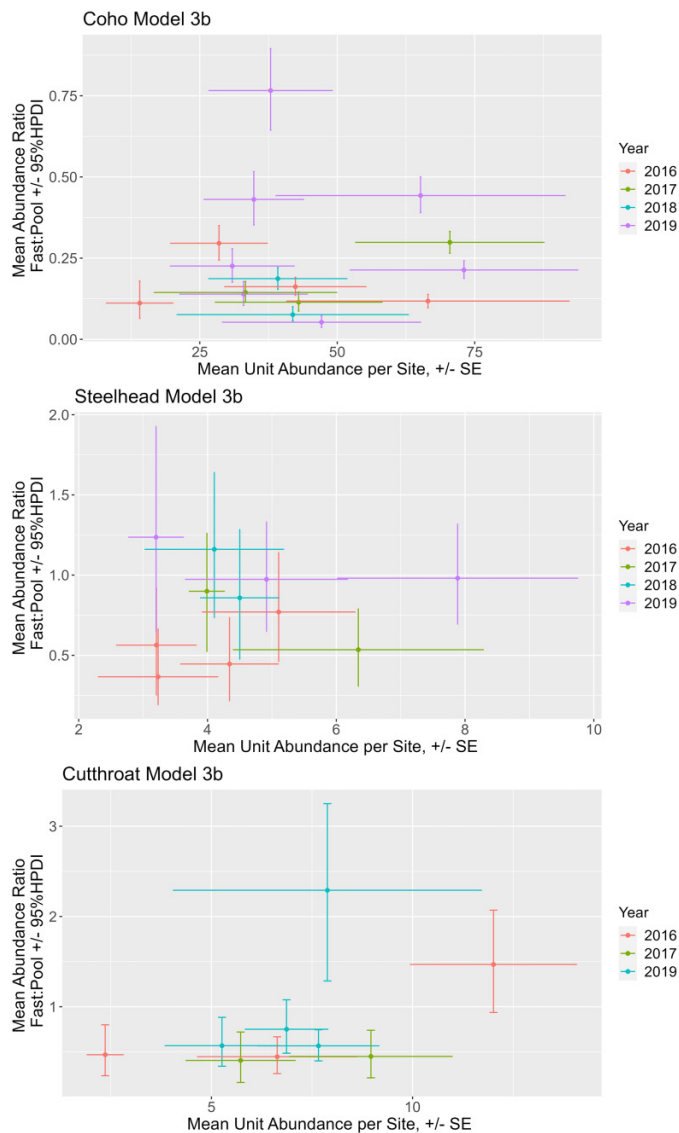
**Fig. 4.** The ratio of abundances in fast-water units relative to pool units for three species (panels) and four years (y-axis). Note difference in scale on the x-axis. Values less than 1 indicate greater abundance in pool units. These distributions include uncertainty within and among sampling units. All estimates were generated with Model 3b.



and fast-water units. We thus did not find a significant signal of density-dependent habitat selection for all three species but note that slopes were negative for all three species.

We used the analytical framework developed above to infer abundance in fast-water units. Using eq. 16, we generated a posterior distribution of the abundance of fish in a pool given a snorkel count of 10 fish (Fig. 6). Application of eq. 20 to a future coho snorkel count in a pool yields abundance in a fast-water unit (Fig. 7). This extrapolation includes uncertainty in proportion seen by snorkelers and hence the associated pool abundance. It also includes uncertainty in the ratio of abundances between pool and fast-water units (3.3x, Fig. 3). Due to extreme uncertainty inferring cutthroat and steelhead abundance in pools, we make no attempt to apply

**Fig. 5.** Search for density-dependent habitat selection. The average unit abundance at a site does not significantly predict the ratio of abundances between pool and fast units within that site. All estimates were generated with model 3b.



eq. 20 and extrapolate abundance to fast-water units for these species.

The proportion of fish seen by snorkelers generally declined with increasing abundance in the pool for coho (Fig. S3) and cutthroat (Fig. S4), but not necessarily steelhead (Fig. S5). We once again note small differences between the alternative mark-recapture model formulations (rows of Figs. S3–S5), and whether or not snorkel counts are integrated into the abundance estimate or not (columns of Figs. S3–S5). The effect of cover and clarity on snorkel detection were not statistically significant for any of the three species, but posteriors for both covariates were negative for all three species (Supplement 4).

## Discussion

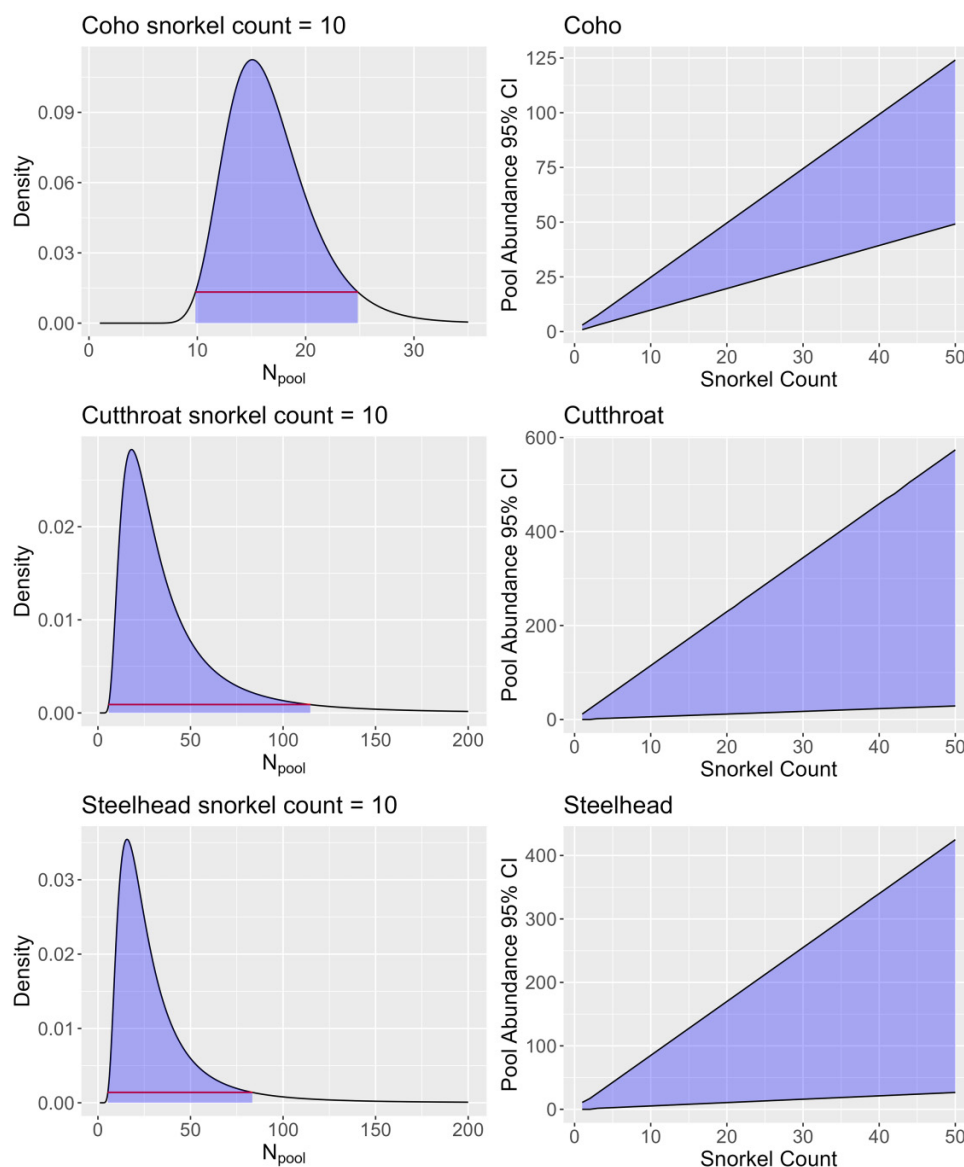
Our analysis of multiple mark-recapture and snorkel counts shares some features of a novel analysis first developed by Staton et al. (2022). Both analyses use Bayesian techniques to integrate uncertainty in mark-recapture estimates into the snorkel detection probability. We further develop estimators for uncertainty in abundance in fast-water units, which may not be snorkeled under all protocols. By quantifying the relative abundance of fish in pool and fast-water units with a fuller accounting of uncertainty, we facilitate estimation across stream unit types and enable the calibration of snorkel counts to population estimates when there is a management need, e.g., management of steelhead fisheries on southwest Oregon rivers (Oregon Department of Fish and Wildlife 2021). In pools, snorkelers observed, on average, 63% of the mark-recapture estimate for coho salmon. This result is very similar to a previous result indicating that snorkelers observed 64% of the mark-recapture estimate (Rodgers et al. 1992), suggesting that snorkeler detection probability has remained consistent over time for OPSW monitoring.

We found a negative effect of abundance on the proportion of juvenile coho and cutthroat seen by snorkelers across all models. Some of the variance in snorkel detection probability can thus be attributed to variance in abundance. However, knowledge of the relationship between abundance and snorkel detection probability has little practical utility because local abundance will generally not be known when a snorkel count is made. In future scenarios when only a snorkel count exists, inference about abundance in the pool should proceed via eqs. 14 and 17 without further attempt to “account” for effects of abundance on the snorkel detection probability. Since cover or clarity had insignificant effects on snorkeler detection probability for all three species, these measures may provide only weak improvement in estimates of abundance derived from snorkel counts.

Density-dependent habitat selection can influence population regulation (Pulliam and Danielson 1991; McPeck et al. 2001), species interactions (Danielson 1992), evolutionary processes (Holt 1987), and the resulting eco-evolutionary dynamics (Morris 2011). Our study was able to detect whether the relative abundance in pools and fast-water units within a site changed with overall abundance in the site. We found no such evidence of density-dependent habitat selection within the range of naturally changing abundances that occurred during our study. However, we note that coho exhibit a strong positive relationship between abundance ratio of units (fast:pool) and the mean unit abundance across years. We do not consider this evidence of density-dependent habitat selection because pooling data across units obscures the spatial scale over which habitat selection occurs (i.e., a site composed of multiple pool and fast-water units). Density-dependent habitat selection of spawning Chinook salmon (*O. tshawytscha*) has been documented in our study region (Falcy 2015), but remains poorly understood for juvenile coho, steelhead, and cutthroat trout.

We focus herein on estimating uncertainty profiles for abundance at the scale of pools and fast-water units, given a pool snorkel count. Managers may desire information at

**Fig. 6.** Probability density for the estimate of the number of juvenile fish of each three species (rows) in a pool given a snorkel count of 10 (left column). The 95% highest probability density interval (HPDI) is given by the red horizontal line and blue shading. The 95% HPDI envelope for pool abundance is given for snorkel counts ranging from 1 to 50 (right column). All estimates generated with Model 3b.

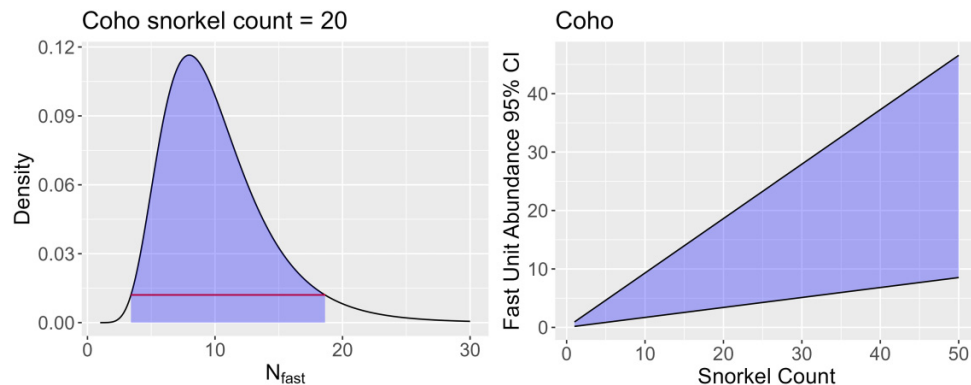


larger spatial scales. For example, a stream reach comprised several pool and fast-water units. The abundance at the scale of the entire stream reach is the sum of abundances in all the pool units (presumably snorkeled) plus the sum of all the abundances in all the fast-water units. The average of the pool snorkel counts could be used in eq. 20 to generate an abundance profile for each fast-water unit in the reach. Let the median of the resulting distribution be the point estimate. If there are four fast-water units, then the point estimate for abundance in all the fast-water units within a stream is simply the point estimate multiplied by 4. Treating each fast-water unit as an independent random variable, we can easily leverage the fact that variances are additive. That is, the variance of the sum of multiple random variables is equal to the sum of the individual variances. Estimating the variance of

the sum of several correlated random variables is also highly tractable: simply add the variances and twice all the pairwise covariances (see Bienaymé's identity). Generating a probability density function for the sum of multiple correlated random variables is less tractable and may require the application of moment generating functions. Regardless, the analytics developed here at the scale of habitat units can be rolled up into an estimate with uncertainty at the scale of an entire stream reach. Estimates at the reach scale can then be used to estimate abundance at larger scales, such as stratum, by extrapolating to the extent of species distribution within the monitored area (Stevens and Olsen 2004).

Monitoring decisions are ideally motivated by a desire to reduce uncertainty around management actions (Nichols and Williams 2006) while also optimizing costs (Falcly 2018). Addi-

**Fig. 7.** Abundance in a fast-water unit is derived from a snorkel count in a pool by accounting for the snorkeler detection probability within and among pool units, and the ratio of abundances between pools and fast-water across all years. The left-hand column illustrates a pool snorkel count of 20 coho. The red horizontal bar and the blue shading give the 95% highest probability density interval (HPDI). The 95% HPDI envelope for fast-water abundance is given for snorkel counts ranging from 1 to 50 (right column). All estimates were generated with Model 3b.



tional work on the efficiency of snorkel surveys could entail simulating management decision-making with uncertainty about biological trends. Our analysis provides the estimation uncertainty that could be used in such a management strategy evaluation (Punt et al. 2016) of an ESA listed species.

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### Notes

The use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government. Fish sampling in this study was done in an ethical manner and in accordance with Oregon Administrative Rule 635-011-0066 and within the confines of ODFW's permitted fisheries research programs (National Marine Fisheries Service permit numbers 19958, 20822, 21768, and 22446).

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## Data availability

Additional figures are given in Supplement A. Raw model outputs are given in Supplement B. Raw data are given in Supplement C.

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### Competing interests

The authors declare there are no competing interests.

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## Supplementary material

Supplementary data are available with the article at <https://doi.org/10.1139/cjfas-2023-0085>.

## References

Bisson, P.A., Reeves, G.H., Bilby, R.E., and Naiman, R.J. 1997. Watershed management and Pacific salmon: desired future conditions. *In* Pacific

- Salmon and their ecosystems: status and future options. Edited by D.J. Stouder, P.A. Bisson and R.J. Naiman. Chapman and Hall, New York. pp. 447–474.
- Busby, P.J., Wainwright, T.C., Bryant, G.J., Lierheimer, L.J., Waples, R.S., Waknitz, F.W., and Agomarsino, I.V. 1996. Status review of West Coast steelhead from Washington, Oregon, and California. NOAA Technical Memorandum NMFS-NWFSC-27, U.S. Department of Commerce.
- Constable, R.J., Jr., and Suring, E. 2022. Juvenile salmonid monitoring in Coastal Oregon and Lower Columbia streams, 2021. Monitoring Program Report Number OPSW-ODFW-2022-1, Oregon Department of Fish and Wildlife, Salem, Oregon.
- Constable, R.J., Jr., and Suring, E. 2023. Implications of metrics and methodology for juvenile salmonid monitoring in Western Oregon streams. *Northwest Sci.* **96**: 63–79. doi:[10.3955/046.096.0106](https://doi.org/10.3955/046.096.0106).
- Crawford, B.A. 2011. Methods for estimating instream juvenile salmonid abundance using snorkeling. Washington Salmon Recovery Funding Board, Olympia, Washington. pp. 41–43.
- Danielson, B.J. 1992. Habitat selection, interspecific interactions and landscape composition. *Evol. Ecol.* **6**: 399–411. doi:[10.1007/BF02270700](https://doi.org/10.1007/BF02270700).
- Dolloff, C.A., Hankin, D.G., and Reeves, G.H. 1993. Basin wide estimates of habitat and fish populations in streams. U.S. Forest Service Southeastern Forest Experiment Station General Technical Report SE-GTR-83. Asheville, North Carolina.
- Ebersole, J.L., Colvin, M.E., Wigington, P.J., Jr., Leibowitz, S.G., Baker, J.P., Church, M.R., et al. 2009. Hierarchical modeling of late-summer weight and summer abundance of juvenile coho salmon across a stream network. *Trans. Am. Fish. Soc.* **138**(5): 1138–1156. doi:[10.1577/T07-245.1](https://doi.org/10.1577/T07-245.1).
- Falcy, M.R. 2015. Density-dependent habitat selection of spawning Chinook salmon: broad-scale evidence and implications. *J. Anim. Ecol.* **84**: 545–553. doi:[10.1111/1365-2656.12297](https://doi.org/10.1111/1365-2656.12297). PMID: [25283166](https://pubmed.ncbi.nlm.nih.gov/25283166/).
- Falcy, M.R. 2018. A cost-optimization framework for planning applied environmental science. *Bioscience*, **68**: 912–922.
- Flitcroft, R., Burnett, K., Snyder, J., Reeves, G., and Ganio, L. 2012. Do network relationships matter? Comparing network and instream habitat variables to explain densities of juvenile coho salmon (*Oncorhynchus kisutch*) in Mid-coastal Oregon, USA. *Aquat. Conserv. Mar. Freshwater Ecosyst.* **22**: 288–302. doi:[10.1002/aqc.2228](https://doi.org/10.1002/aqc.2228).
- Flitcroft, R., Burnett, K., Snyder, J., Reeves, G., and Ganio, L. 2014. River-scape patterns among years of juvenile coho salmon (*Oncorhynchus kisutch*) in Mid-coastal Oregon: implications for conservation. *Trans. Am. Fish. Soc.* **143**: 26–38. doi:[10.1080/00028487.2013.824923](https://doi.org/10.1080/00028487.2013.824923).
- Gardiner, W.R. 1984. Estimating population densities of salmonids in deep water in streams. *J. Fish Biol.* **24**: 41–49. doi:[10.1111/j.1095-8649.1984.tb04774.x](https://doi.org/10.1111/j.1095-8649.1984.tb04774.x).
- Gelman, A., Hwang, J., and Vehtari, A. 2014. Understanding predictive information criteria for Bayesian models. *Stat. Comput.* **24**: 997–1016. doi:[10.1007/s11222-013-9416-2](https://doi.org/10.1007/s11222-013-9416-2).
- Gelman, A., Meng, X., and Stern, H.S. 1996. Posterior predictive assessment of model fitness via realized discrepancies (with discussion). *Stat. Sin.* **6**: 733–807.
- Gelman, A., and Rubin, D.B. 1992. Inference from iterative simulation using multiple sequences. *Stat. Sci.* **7**: 457–511. doi:[10.1214/ss/1177011136](https://doi.org/10.1214/ss/1177011136).
- Hankin, D.G., and Reeves, G.H. 1988. Estimating total fish abundance and total habitat area in small streams based on visual estimation methods. *Can. J. Fish. Aquat. Sci.* **45**: 834–844. doi:[10.1139/f88-101](https://doi.org/10.1139/f88-101).
- Hawkins, D.K. 1997. Hybridization between coastal cutthroat (*Oncorhynchus clarki clarki*) and steelhead trout (*O. mykiss*). Doctoral dissertation, University of Washington, Seattle.
- Hayes, J.W., and Baird, D.B. 1994. Estimating relative abundance of juvenile brown trout in rivers by underwater census and electrofishing. *N.Z. J. Mar. Freshwater Res.* **28**: 243–253. doi:[10.1080/00288330.1994.9516612](https://doi.org/10.1080/00288330.1994.9516612).
- Hicks, B.J., and Watson, N.R.N. 1985. Seasonal changes in abundance of brown trout (*Salmo trutta*) and rainbow trout (*S. gairdnerii*) assessed by drift diving in the Rangitikei River, New Zealand. *N.Z. J. Mar. Freshwater Res.* **19**: 1–10. doi:[10.1080/00288330.1985.9516070](https://doi.org/10.1080/00288330.1985.9516070).
- Hicks, B.J., and Hall, J.D. 2003. Rock type and channel gradient structure Salmonid populations in the Oregon coast range. *Trans. Am. Fish. Soc.* **132**: 468–482, 2003. doi:[10.1577/1548-8659\(2003\)132%3c0468:RTACGS%3e2.0.CO;2](https://doi.org/10.1577/1548-8659(2003)132%3c0468:RTACGS%3e2.0.CO;2).
- Hillman, T.W., Mullan, J.W., and Griffith, J.S. 1992. Accuracy of underwater counts of juvenile Chinook salmon, coho salmon, and steelhead trout. *North Am. J. Fish. Manage.* **12**: 598–603. doi:[10.1577/1548-8675\(1992\)012%3c0598:AOUCOJ%3e2.3.CO;2](https://doi.org/10.1577/1548-8675(1992)012%3c0598:AOUCOJ%3e2.3.CO;2).
- Holt, R.D. 1987. Population dynamics and evolutionary processes: the manifold roles of habitat selection. *Evol. Ecol.* **1**: 331–347. doi:[10.1007/BF02071557](https://doi.org/10.1007/BF02071557).
- Huggins, R.M. 1991. Some practical aspects of conditional likelihood approach to capture experiments. *Biometrics*, **47**: 725–732. doi:[10.2307/2532158](https://doi.org/10.2307/2532158).
- Jonasson, B.C., Sedell, E.R., Tattam, S.K., Garner, A.B., Horn, C., Bliesner, K.L., et al. 2016. Investigations into the life history of naturally produced spring Chinook salmon and summer steelhead in the Grande Ronde River subbasin. Annual Report BPA Project #1992-026-04, Oregon Department of Fish and Wildlife, LaGrande, OR.
- Kéry, M., and Schaub, M. 2012. Bayesian population analysis using WinBUGS: a hierarchical perspective. Academic Press.
- Korman, J., Decker, S.A., Mossop, B., and Hagen, J. 2010. Comparison of electrofishing and snorkeling mark-recapture estimation of detection probability and abundance of juvenile steelhead in a medium sized river. *North Am. J. Fish. Manage.* **30**(5): 1280–1302. doi:[10.1577/M09-159.1](https://doi.org/10.1577/M09-159.1).
- May, C.L., and Lee, D.C. 2004. The relationships among in-channel sediment storage, pool depth, and summer survival of juvenile salmonids in Oregon coast range streams. *North Am. J. Fish. Manage.* **24**: 761–774. doi:[10.1577/M03-073.1](https://doi.org/10.1577/M03-073.1).
- McElhany, P., Ruckelshaus, M.H., Ford, M.J., Wainwright, T.C., and Bjorkstedt, E.P. 2000. Viable salmonid populations and the recovery of evolutionarily significant units. U.S. Department of Commerce, NOAA Technical Memorandum NMFS-NWFSC-42.
- McPeck, M.A., Rodenhouse, N.L., Holmes, R.T., and Sherry, T.W. 2001. A general model of site-dependent population regulation: population-level regulation without individual-level interactions. *Oikos*, **94**: 417–424. doi:[10.1034/j.1600-0706.2001.940304.x](https://doi.org/10.1034/j.1600-0706.2001.940304.x).
- Moore, K.M.S., Jones, K.K., and Dambacher, J.M. 1997. Methods for stream habitat surveys. Oregon Department of Fish and Wildlife, Information Report 97-4, Portland, Oregon.
- Morris, D.W. 2011. Adaptation and habitat selection in the evolutionary process. *Proc. Royal Soc. B*, **278**: 2401–2411. doi:[10.1098/rspb.2011.0604](https://doi.org/10.1098/rspb.2011.0604). PMID: [21613295](https://pubmed.ncbi.nlm.nih.gov/21613295/).
- Mullner, S.A., Hubert, W.A., and Wesche, T.A. 1998. Snorkeling as an alternative to depletion electrofishing for estimating abundance and length-class frequencies of trout in small streams. *North Am. J. Fish. Manage.* **18**: 947–953. doi:[10.1577/1548-8675\(1998\)018%3c0947:SAATD%3e2.0.CO;2](https://doi.org/10.1577/1548-8675(1998)018%3c0947:SAATD%3e2.0.CO;2).
- Nichols, J.D., and Williams, B.K. 2006. Monitoring for conservation. *Trends Ecol. Evol.* **21**: 668–673. doi:[10.1016/j.tree.2006.08.007](https://doi.org/10.1016/j.tree.2006.08.007). PMID: [16919361](https://pubmed.ncbi.nlm.nih.gov/16919361/).
- Nickelson, T.E., Rodgers, J.D., Johnson, S.L., and Solazzi, M.F. 1992. Seasonal changes in habitat use by juvenile coho salmon (*Oncorhynchus kisutch*) in Oregon coastal streams. *Can. J. Fish. Aquat. Sci.* **49**: 783–789. doi:[10.1139/f92-088](https://doi.org/10.1139/f92-088).
- O'Neal, J.S. 2007. Snorkel surveys. In *Salmonid field protocols handbook: techniques for assessing status and trends in Salmon and Trout populations*. Edited by D.H. Johnson, B.M. Shrier, J.S. O'Neal, J.A. Knutzen, X. Augerot, T.A. O'Neal and T.N. Pearsons. American Fisheries Society, Bethesda, Maryland. pp. 325–340.
- ODFW, 2021. Rogue-South Coast Multi-Species Conservation and Management Plan. [https://www.dfw.state.or.us/fish/CRP/docs/rogue\\_south\\_coast\\_multispecies/RSP\\_main\\_final.pdf](https://www.dfw.state.or.us/fish/CRP/docs/rogue_south_coast_multispecies/RSP_main_final.pdf).
- Peterson, J.T., Thurow, R.F., and Guzevich, J.W. 2004. An evaluation of multipass electrofishing for estimating the abundance of stream-dwelling salmonids. *Trans. Am. Fish. Soc.* **133**: 462–475. doi:[10.1577/03-044](https://doi.org/10.1577/03-044).
- Peterson, J.T., Banish, N.P., and Thurow, R.F. 2005. Are block nets necessary? Movement of stream dwelling salmonids in response to three common survey methods. *North Am. J. Fish. Manage.* **25**: 732–743. doi:[10.1577/M04-051.1](https://doi.org/10.1577/M04-051.1).
- Plummer, M. 2003. JAGS: a program for analysis of Bayesian graphical models using Gibbs sampling. In *Proceedings of the 3rd International Workshop on Distributed Statistical Computing*, Vienna, March 20–22, 2003.

Poole, J.R., Barnett, B., Brown, E.T., Copeland, T., McClure, C., Putnam, S., et al. 2019. Idaho anadromous parr monitoring. Annual report 2019. Idaho Department of Fish and Game Report 20-04, Boise, ID.

Pulliam, R.H. and Danielson, B.J. 1991. Sources, sink and habitat selection: a landscape perspective on population dynamics. *Am. Nat.* **137**: S50–S66. doi:10.1086/285139.

Punt, A.E., Butterworth, D.S., de Moor, C.L., De Oliveira, J.A.A., and Had- don, M. 2016. Management strategy evaluation: best practices. *Fish. Fish.* **17**: 303–334. doi:10.1111/faf.12104.

R Core Team. 2023. R: A language and environment for statistical comput- ing. R Foundation for Statistical Computing, Vienna, Austria. Avail- able from <https://www.R-project.org/>.

Rodgers, J.D., Solazzi, M.F., Johnson, S.L., and Buckman, M.A. 1992. Com- parison of three techniques to estimate juvenile coho salmon popu- lations in small streams. *North Am. J. Fish. Manage.* **12**: 79–86. doi:10.1577/1548-8675(1992)012%3c0079:COTTTE%3e2.3.CO;2.

Rodgers, J.D. 2000. Abundance of juvenile coho salmon in Oregon coastal streams, 1998 and 1999. Monitoring Program Report Number OPSW- ODFW-2000-1, Oregon Department of Fish and Wildlife, Portland, Oregon.

Roni, P., and Fayram, A. 2000. Estimating winter salmonid abun- dance in small Western Washington streams: a comparison of three techniques. *North Am. J. Fish. Manage.* **20**: 682–691. doi:10.1577/ 1548-8675(2000)020%3c0683:EWSAIS%3e2.3.CO;2.

Rosenberger, A.E., and Dunham, J.B. 2005. Validation of abundance esti- mates from mark-recapture and removal techniques for rainbow trout captured by electrofishing in small streams. *North Am. J. Fish. Manage.* **25**: 1395–1410. doi:10.1577/M04-081.1.

Royle, A., Dorazio, R.M., and Link, W.A. 2007. Analysis of multinomial models with unknown index using data augmentation. *J. Comput. Graph. Stat.* **16**: 67–85. doi:10.1198/106186007X181425.

Satterthwaite, T.D. 2002. Klamath Mountains Province Steelhead trout project. 1999 Annual report. Monitoring Program Report Number OPSW-ODFW-2002-09, Oregon Department of Fish and Wildlife, Port- land.

Schaub, M., and Abadi, F. 2011. Integrated population models: a novel analysis framework for deeper insights into population dynamics. *J. Ornithol.* **152**: S227–S237. doi:10.1007/s10336-010-0632-7.

Schill, D.J., and Griffith, J.S. 1984. Use of underwater observations to esti- mate cutthroat trout abundance in the Yellowstone River. *North Am. J. Fish. Manage.* **4**: 479–487. doi:10.1577/1548-8659(1984)4%3c479: UOUOTE%3e2.0.CO;2.

Spiegelhalter, D.J., Best, N.G., Carlin, B.P., and Van Der Linde, A. 2002. Bayesian measures of model complexity and fit. *Stat. Methodol.* **Ser. B**, **64**: 583–639. doi:10.1111/1467-9868.00353.

Staton, B.A., Justice, C., White, S., Sedell, E.R., Burns, L.A., and Kaylor, M.J. 2022. Accounting for uncertainty when estimating drivers of imper- fect detection: an integrated approach illustrated with snorkel sur- veys for riverine fishes. *Fish. Res.* **249**: 106209. doi:10.1016/j.fishres. 2021.106209.

State of Oregon. 1997. The Oregon plan (Oregon Coastal Salmon Restora- tion Initiative). Oregon Governor's Office. Salem, Oregon.

Stevens, D.L., Jr., and Olsen, A.R. 2004. Spatially-balanced sampling of natural resources. *J. Am. Stat. Assoc.* **99**(465): 262–278. doi:10.1198/ 016214504000000250.

Stout, H.A., Lawson, P.W., Bottom, D.L., Cooney, T.D., Ford, M.J., Jordan, C.E., et al. 2012. Scientific conclusions of the status review for Oregon coast coho salmon (*Oncorhynchus kisutch*). U.S. Dept. Commer. NOAA Tech. Memo. NMFS-NWFSC-118.

Temple, G.M., and Pearsons, T.N. 2007. Electrofishing: backpack and drift boat. In *Salmonid field protocols handbook: techniques for assess- ing status and trends in Salmon and Trout populations*. Edited by D.H. Johnson, B.M. Shrier, J.S. O'Neal, J.A. Knutzen, X. Augerot, T.A. O'Neal and T.N. Pearsons. American Fisheries Society, Bethesda, Maryland. pp. 95–132.

Thurrow, R.F. 1994. Underwater methods for study of salmonids in the In- termountain West. U.S. Forest Service, Intermountain Research Sta- tion, General Technical Report INT-GTR-307, Ogden, Utah.

Tualatin River Watershed Council. 2015. Tualatin river rapid bio- assessment 2013 & 2014 final report. Prepared by Bio Surveys, LLC, Portland, Oregon.

Wainwright, T.C., Chilcote, M.W., Lawson, P.W., Nickelson, T.E., Hunt- ington, C.W., Mills, J.S., et al. 2008. Biological recovery crite- ria for the Oregon coast coho salmon evolutionarily significant unit. U.S. Dept. Commerce, NOAA Technical. Memorandum. NWFS- NWFSC-91.

Walters, J., Hansen, J., Lockhart, J., Reighn, C., Keith, R., and Olson, J. 1999. Idaho supplementation studies: five year report 1992-1996. Idaho Department of Fish and Game Report 99-14. Project 89-098. Bonneville Power Administration, Portland, Oregon.

## Appendix A

Using the R programming language, define shape and rate parameters for *RatioFP* and *PS*:

```
shape1<-14.00224 #(shape on RatioFP; numerator)
shape2<-12.0917 #(shape on PS; denominator)
rate1<-47.79727 #(rate on RatioFP; numerator)
rate2<-19.13105 #(rate on PS; denominator)
```

Simulate observations

```
RatioFP<-rgamma(100000, shape1, rate1)
PS<-rgamma(10000, shape2, rate2)
SC<-20
Obs.sim<-(SC/PS)*RatioFP
```

Calculate the probability density

```
fast<-seq(1,30, by = 0.001)
Obs.calc<-dbetapr(fast/20, shape1, shape2,
scale = rate2/rate1)*abs(1/20)
```

Compare *Obs.sim* and *Obs.calc* in Fig. A1 for a demonstra- tion of concordance between the derived mathematical func- tion and the simulation.

We further note that our closed-form posterior numerically integrates to 1 with absolute error <1.4E-06

```
f<-function(x){dbetapr(x/20, shape1, shape2,
scale = rate2/rate1)*abs(1/20)}
integrate(f,0, Inf)
```

**Fig. A1.** Concordance between simulated and calculated functions for the abundance of fish in a pool unit.

